Analysis of existing heuristics and implementing a new obfuscation method for hiding influential individuals in community networks

Dissertation Report

M.Sc. Data Science

by

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Declaration:

I affirm that I have identified all my sources and that no part of my dissertation paper uses unacknowledged materials.

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Abstract: In today's age people across the world are heavily connected through social media platforms. Many social media circles are led by one or more influential members. Such members are labeled as 'masterminds'. They are responsible for organizing major events with the help of its community members and sometimes they require privacy or concealment to effectively carry out these events. We analyze and review this concealment problem as stated by some researchers and also suggest some new heuristics to improve on the existing algorithms. Centrality measures and metrics like IC, LT have been used to measure the effectiveness of these algorithms.

Keywords: Network Analysis, Hiding Heuristic, Independent Cascade(IC), Linear Threshold(LT), Community Detection, Centrality measures.

1 Introduction

Network analysis has been a crucial research topic across many domains in different capacities for years. In 21st century, network analysis has gained even more importance in terms of research scopes. Today's world is not just based on individual systems but also based on connected systems to a great extent. Analysing such connected systems helps us infer a great deal of things which can be used to impact the society in many positive ways. Social media networks like Facebook, X, Instagram generate massive amount of data on a daily basis. Network analysis can help us understand viral content propagation, fake news diffusion, community clustering, opinion polarization etc. These are critical in ensuring election security, public policy and marketing. In epidemiology and pandemic modeling, network analysis can help in contact tracing, locating infection transmission pathways and thus forming vaccination strategies. Network analysis is of utmost importance when it comes to ensuring national security where terrorist communication networks are needed to be studied. Cyber attack pathways can also be tracked via network analysis. Intelligence agencies often use network analysis for such purposes. In banks and fintech firms, network based models are used to detect money laundering rings and fraudulent transaction clusters. Network analysis helps study systemic risk in financial institutions.

Basic knowledge of graph theory forms the backbone of all these researches. In many such networks, it becomes important to correctly identify the influential nodes or members. For example, in

biological networks, knowing the key genes, key metabolites are important to understand health regulation and disease propagation. In terrorist networks, it becomes important for the counter-terrorism teams to identify the mastermind in the terrorist network. No specific algorithm can be used as a general solution for such network analysis. Heuristics have been the way forward. Some state of the art heuristic algorithms with their respective advantages and limitations have been stated and published by many researchers across the globe. Our focus in this work is on social networks like Facebook and some crime networks. Datasets of such networks are available in reputed websites like SNAP.

The purpose of our problem is two fold. One part of the problem is about identification of the influential nodes in the network (for eg from the point of view of crime investigators in crime networks). Another part of the problem is from the point of view of credible users of a product in a network or can also be seen from the point of view of counter terrorism units. For eg privacy is an important concern for all legitimate users of social media. For counter-terrorism teams, hiding their own important nodes(eg ethical hackers) from cyber attackers are an important concern as well. Researchers apply detection heuristic on a network to identify the influential nodes and later apply hiding heuristic on such nodes. The detection heuristic judges how fast it is being able to detect the influential node and the hiding heuristic judges how well it is being able to hide the influential nodes from exposure while at the same time trying to maintain their influence in the network. In a way, it is an optimization problem i.e. hiding yet maintaining their network influence on the lesser important nodes. Based on the respective purpose, the application would be different in different networks but the two fold objective is conducted and tested on the same network at a time.

Commonly known centrality measures like degree centrality, closeness centrality and betweenness centrality have been used to measure the importance of nodes in the network. To evaluate how well the influential nodes are being able to influence the network after being concealed, we use two metrics, namely - IC and LT. The IC (Independent Cascade) metric captures the ease with which the mastermind's influence can probabilistically diffuse through the network. The LT (Linear Threshold) metric captures the structural resistance of the network to influence diffusion originating from the mastermind.

Different researchers have attacked the problem of hiding from different angles. Some have chosen to hide the mastermind from a global perspective of the network(eg: ROAM heuristic). Some others have divided the network into smaller communities using community detection algorithms, followed by identifying a core community and then modified connections flowing from within the core community to the less important communities(eg HERMES, DICE Heuristics). In our suggestion of a new heuristic, we have approached the problem from the latter angle. We have tried to ideate an improved heuristic, develop the code and compare the results of our heuristic with the existing ones.

2 Problem Description

In the current age of datafication, personal data in social networks are often at risk of being exposed. Almost the entire global population is connected to each other via social media platforms. Governments and corporates often operate within grey areas of privacy rules and try to manipulate regulations in a way which benefits their purposes and profits. Till date, many analytical techniques have been devised with an aim to uncover important informations from social media networks, one of which is the identification and detection of influential individuals within a social circle. However, with the rise in the usage of these advanced analytics tools, the concerns related to privacy and ethics have also heightened over time. For example, Mislove et al.[9] have shown how, by analysing Facebook's social network structure and the features and characteristics of some users, important personal informations about other users can also be inferred.

To handle such privacy issues, various measures have been thought of, like ensuring legal controls through algorithmic implementations and developing market-like mechanisms which allow individual users to monetize their personal data. However, such strategies have only remained at a thought level and not been implemented properly on a governmental scale. In fact, on the contrary, in the recent times we have seen Facebook's release of "Global Government Requests Report"[1] which revealed a global increase in government requests to secretly access user data. Furthermore, in dictatorial regimes, social networking sites are heavily policed and anti-establishment contents are often censored. Hence the issue of privacy still remains largely unresolved [6] [7].

Against this backdrop, some researchers have asked the question: can the users or communities

proactively take steps to manage their social connections in a way that their privacy is less exposed to the workings of the network analysis tools ? i.e. can they disguise their presence in the network to escape the detection mechanisms ? While focus has mainly been on developing even better network analysis tools, this basic question has received little importance in the literature so far. This is an important question to address as it helps the social media users from getting their data encroached by the government and corporates. Also, on the other hand, this study assists counter-terrorism and intelligence teams to understand how terrorists and criminals try to evade detection techniques in social networks given their increasing reliance on social media platforms in the recent times[10][5].

The question raised above can be addressed by taking into account the three main centrality measures namely degree, closeness and betweenness centrality which are generally used to identify an user's importance in the network. The aim for the users shall be to avoid being spotlighted by these three measures while preserving their influence as much as possible. From a graph theoretic perspective, this is basically an optimization problem. We must note its computational complexity so that we can reflect on the theoretical limits of such an act as disguising oneself. Though an optimal solution, as has been shown in literature, is often hard to compute yet we have seen how different researchers have demonstrated the effectiveness of some suprisingly simple heuristics, whereby the rewiring of social connections in the individual's immediate neighbourhood has paid off amazing results. The rewiring involves two actions that are already feasible in popular social media platforms: (i) "unfriending" certain neighbours (ii) introducing certain neighbours to each other[13].

In some heuristics, we shall try to approach the problem from a group's point of view where we shall see how a community can conceal itself to increase the likelihood of being overlooked by community detection algorithms. We shall introduce a term called "modularity" as a measure of concealment designed to quantify the degree to which a community is hidden.

3 Centrality Measures

Network centrality measures helps us to quantify the importance of any given node in the network. They can thus help us to identify the most important nodes in a graph. The most important nodes are the most influential nodes in a social network i.e. nodes which can effectively propagate information to many other nodes in the network. The centrality measures also helps us identify smaller core

communities from a large community structure. The most commonly known centrality measures are degree, closeness and betweenness centralities.

Degree centrality (as per Shaw 1954) says that the importance of a node is proportional to the number of its neighbors[12] i.e. the degree centrality of a node v in network G is :

$$C_{degr}(G, v) = |N_G(v)| \tag{1}$$

Here, $N_G(v)$ denotes the neighborhood of node v in graph G. This is the set of all nodes directly connected to v by an edge. $|N_G(v)|$ denotes the cardinality (size) of that set, i.e., the number of neighbors node v has.

Closeness centrality (as per Beauchamp 1965) measures the importance of a node in terms of shortest distances from this node to all other nodes in the network[12]. Mathematically, the closeness centrality of a node v in network G can be expressed as :

$$c_{clos}(G, v) = \sum_{w \in V \setminus \{v\}} \frac{1}{\lambda_G(v, w)} \tag{2}$$

Here, V denotes the set of all nodes in the network G, w denotes a node in the network (other than v) from which we measure the distance of v. $V \setminus \{v\}$ denotes all nodes in the network except v itself. $\lambda_G(v, w)$ denotes the length of the shortest path between node v and node w in network G. If no path exists, this distance is considered infinite. The reciprocal of the shortest path length $\frac{1}{\lambda_G(v,w)}$ indicates that closer nodes (i.e. shorter paths) contribute more to the centrality score, while distant nodes contribute less. The summation adds up these reciprocal distances for all other nodes in the network. So, closeness centrality measures how close a node is to all others in the network i.e. the smaller the average distance to other nodes, the higher the closeness centrality.

Betweenness centrality (Anthonisse 1971; Freeman 1977) states that, if we take into account all the shortest paths in a network, then the more such paths pass through a given node, the more important the role of that node in the entire network i.e. the node controls such paths[12]. Mathematically, the

betweenness centrality of node $v \in V$ in network G is given by :

$$c_{betw}(G, v) = \sum_{w,u \in V \setminus \{v\}} \frac{|\{p \in \pi_G(w, u) : v \in p\}|}{|\pi_G(w, u)|} \tag{3}$$

v is the specific node whose betweenness centrality we want to measure. $w, u \in V \setminus \{v\}$ denotes all pairs of distinct nodes in the network excluding v itself. $\pi_G(w, u)$ is the set of all shortest paths between nodes w and u in graph G. $|\pi_G(w, u)|$ is the total number of shortest paths between w and u. $\{p \in \pi_G(w, u) : v \in p\}$ is the subset of those shortest paths that pass through node v. $|\{p \in \pi_G(w, u) : v \in p\}|$ is the number of shortest paths between w and u that include v. The fraction is the proportion of shortest paths between w and u that go through v.

So, betweenness centrality measures how often a node lies between other nodes on their shortest paths. A node with high betweenness centrality acts like a bridge or broker in the network, controlling information flow.

4 Objectives

4.1 First Objective

Given a network and a "source node" (or "influential" node, also called as "mastermind" node) $v^\dagger$, the objective is to conceal the importance of $v^\dagger$ by decreasing its centrality measures without compromising its influence over the network. It is achieved by rewiring the links of the network, without exceeding a certain budget. The budget refers to the maximum number of links allowed to be modified i.e. added or removed in the entire rewiring process. This rewiring mechanism performed to disguise $v^\dagger$ can be divided into two consecutive phases :

(i) Part of the budget is spent on minimizing the three centrality measures. In this phase, the influence of $v^\dagger$ shall decrease. This is called the "centrality minimization problem".

(ii) In the second phase, the remaining budget is used to rewire the network in a way so that the influence lost by $v^\dagger$ in the first phase can be recovered as much as possible. It must be ensured that no such links can be added between nodes which were removed in the first phase.

In the literature, two variants of the influence recovery problem as stated in phase (ii) is considered. These are (a) the "individual influence recovery" problem, where the objective is to recover the influence of $v^\dagger$ on every single node (b) the "global influence recovery" problem where the goal is to recover the sum of influences of $v^\dagger$ over all the nodes[13].

4.2 Second Objective

Given a community i.e. a subset of nodes, say, $C^\dagger$, the objective is to conceal the identity of $C^\dagger$ by hiding its existence within the network structure where it exists. A community network structure is a partition of the set of nodes into disjoint and exhaustive subsets or communities. $C^\dagger$ is considered exposed if a community detection algorithm is able to return a community structure CS such that $C^\dagger \in CS$. Again, hiding $C^\dagger$ takes place by rewiring certain links of the network according to some budget which is the maximum number of allowed modifications[13].

5 Models of Influence

In relation to the objective of influence preservation, we bring into the picture the two most prevalent diffusion models in computer science, namely - the independent cascade and the linear threshold model. We shall describe the properties of these models and show how those properties help in addressing and quantifying the problem of influence recovery.

5.1 Linear Threshold Model

In the linear threshold model, every directed edge between two nodes say v and w has a fixed weight $b_{v,w}$. This weight reflects the strength of the influence of node w over node v. We have the following setup for the linear threshold model [3] :

- A node v has a random threshold $\theta_v \sim U[0, 1]$
- A node v is influenced by each neighbour w according to a weight $b_{v,w}$ such that,

$$\sum_{w \text{ neighbor of } v} b_{v,w} \leq 1$$

- A node v becomes active when the sum of the weights from its active neighbours becomes greater than the internal threshold of node v i.e.

$$\sum_{w \text{ active neighbor of } v} b_{v,w} \geq \theta_v$$

5.2 Independent Cascade Model

In this model, the modeling of influences(activation) of nodes is based on probabilities in a directed graph. Given a directed finite graph $G = (V, E)$ and an active subset of nodes S, this model has the following characteristics:

- Each edge (v, w) has a probability p_{vw}.
- If node v becomes active, it gets one chance to make w active with probability p_{vw}.
- An active node gets a single chance to activate its direct neighbours. The activation stops when all the active nodes have exhausted their chances of activating their neighbours.
- If u and v are both active and are having edges connected to w then it does not matter who tries to activate w first.

In context of both these models, an "active" node can be imagined as an infected person who tries to influence but might not always be successful in infecting its neighbours. Analogously, an "inactive" node can be a healthy person who might be influenced by some of its infected neighbours. The stronger the influence, the stronger are the chances of infection[13][3].

Note: Both these models start with an active subset of nodes called the "seed" set. With time passing, in discrete rounds, new nodes get activated due to influence from previously activated nodes. The two models vary in the manner influence spreads through the network. In the independent cascade model, an active node activates each of its neighbours with some pre-defined probability. A random number is generated from a $Unif(0, 1)$. If the random number is less than the probability value, the node gets activated otherwise it stays inactive. In the linear threshold model, each node has some random pre-defined threshold and gets activated when the summation of the weights or influences from its neighbours exceeds that threshold.

6 Hardness Information

In computer science, different kinds of problems exist based on the level of difficulty of the problem. They are hence classified on the basis of complexity. The different kinds of problems are: (i) Problems which cannot be solved by computers (ii) Problems which can be solved by computers efficiently (iii) Problems which have no efficient solution.

P (or polynomial time) problems are easy for computers to solve. A Problem is in P if it can be solved quickly in polynomial time as in n, n^2, n^3 etc. A problem is in NP(Non deterministic polynomial time) class problem if the solution to the problem is easy to verify but not always easy to solve. A problem is called NP-Hard if it is atleast as hard to find a solution as NP problems. They may not even be in NP i.e. verifying may also be hard. The hardest problems inside NP are the NP-complete problems(NPC). They are in NP(can be verified quickly) and they are as hard as any NP problem.

In our context, finding an optimal process to hide a node's importance inside a social network is very difficult from a computation point of view since it would require knowledge of the entire network structure and may also require adding or removing links that are far from the mastermind node. Disguising degree centrality is a P problem whereas disguising closeness and betweenness centrality are NPC problems. Individual influence recovery and global influence recovery are also NPC problems[13].

6.1 Disguising Centrality

The problem of disguising centrality is defined by a tuple, $(G, v^\dagger, b, c, \hat{R}, \hat{A})$, where $G = (V, E) \in \mathcal{G}$ is a network, $v^\dagger \in V$ is the source node or mastermind node (whose centrality is to be minimized[13]), $b \in \mathbb{N}$ is a budget specifying the maximum number of edges that can be added or removed, $c : \mathcal{G} \times V \to \mathbb{R}$ is a centrality measure, $\hat{R} \subseteq E$ is a set of edges whose removal is forbidden, $\hat{A} \subseteq (V \times V) \setminus E$ is a set of edges whose addition is forbidden. The goal is then to identify two sets of edges, $R^* \subseteq (E \setminus \hat{R})$ and $A^* \subseteq (V \times V) \setminus (E \cup \hat{A})$, such that: $|A^*| + |R^*| \leq b$ and $G^* = (V, (E \cup A^*) \setminus R^*)$ is connected (strongly connected if G is directed) and G^* is in:

$$\underset{G' \in \{(V,(E \cup A) \setminus R) : R \subseteq (E \setminus \hat{R}),\ A \subseteq (V \times V) \setminus (E \cup \hat{A})\}}{\arg\min} c(G', v^\dagger).$$

6.2 Individual Influence recovery

This problem is defined by a tuple, $(G, v^\dagger, inf, \hat{A}, f)$, where $G = (V, E) \in \mathcal{G}$ is a network, $v^\dagger \in V$ is the source node or mastermind node (whose influence is to be recovered[13]), $inf : V \times V \to \mathbb{R}$ is an influence measure, $\hat{A} \subseteq (V \times V) \setminus E$ is a set of edges whose addition is forbidden, and $f : V \to \mathbb{R}$ specifies the influences to be recovered (i.e., for every $v_i \in V$ we want the influence of $v^\dagger$ over v_i to be at least $f(v_i)$). The goal is then to identify a set of edges, A^*, that is in:

$$\underset{A \subseteq (V \times V) \setminus (E \cup \hat{A}) : \forall v_i \in V \ \inf_{(v, E \cup A)}(v^\dagger, v_i) \geq f(v_i)}{\arg\min} |A|.$$

6.3 Global Influence Recovery

This problem is defined by a tuple, $(G, v^\dagger, inf, \hat{A}, \phi)$, where $G = (V, E) \in \mathcal{G}$ is a network, $v^\dagger \in V$ is the source node (whose influence is to be recovered[13]), $inf : V \times V \to \mathbb{R}$ is an influence measure, $\hat{A} \subseteq (V \times V) \setminus E$ is a set of edges whose addition is forbidden and $\phi \in \mathbb{R}$ is the total influence to be recovered. The goal is then to identify a set of edges, A^*, that is in:

$$\arg \underset{A \subseteq (V \times V) \setminus (E \cup \hat{A}) \ : \ \inf(v, E \cup A)(v^\dagger) \geq \phi}{\min} |A|.$$

7 Heuristics

The word heuristics comes from the greek work “heuriskein” which means “to find”. In the history of Science, numerous attempts have been made to formulate methods for finding proofs and for arriving at new discoveries. This was earlier famously called as “the art of discovery” and later as “heuristics”[4]. A heuristic is an experience based technique for problem solving that ends up fetching a satisfactory solution quickly when finding an optimal solution to the problem is very difficult. It does not always find the best solution but always finds a very good one. It may not as well produce the best solution in the worst cases but worst cases rarely arise in real life. In the following subsections, some popular state of the art heuristics like - ROAM, DICE, HERMES shall be discussed and analysed below following which our newly implemented heuristic named C3R shall also be discussed in relation to the problem of hiding an influential member in a social network.

Note: In a social network, it is typically seen that one has very limited knowledge of the social connections beyond his or her immediate contacts or at the most mutual connections. Considering one was able to accumulate information about the entire network structure, hardness information that we discussed earlier suggests that it is highly unlikely for an individual to have the required computation power to optimally disguise himself or herself. Against this backdrop, researchers have so far investigated the possibility of disguising an individual's centrality while restricting one's attention to only his or her immediate neighbourhood[13]. Also, it has been kept in mind that massive computational powers or knowledge in sophisticated optimization techniques are not required. The heuristics that have been discussed below are simple enough for an average user of social networking platforms to understand and implement.

7.1 ROAM - Remove One, Add Many

Given a budget b, the ROAM heuristic involves the following steps[13]:

Step 1: Remove the link between the source node("mastermind"), $v\dagger$ and its neighbour of choice v_0.

Step 2: Connect v_0 to $b-1$ nodes of choice, who are neighbours of $v\dagger$ but not of v_0 (if there are fewer than $b-1$ such neighbours, connect v_0 to all of them).

Comments on the heuristic: In order to reduce the centrality of $v\dagger$, **Step 1** is clearly beneficial. Cutting off $v\dagger$ from one of its neighbours will end up reducing the degree centrality of $v\dagger$. Similarly, **Step 1** can decrease the closeness centrality of $v\dagger$ if shortest paths between $v\dagger$ and some other node run through the removed link. **Step 1** can also decrease the betweenness of $v\dagger$ if some of the shortest paths going through $v\dagger$ contain the removed link. However this step may be harmful in depriving $v\dagger$ from its direct influence over v_0.

Step 2 is designed to compensate for any influence lost by $v\dagger$ during **Step 1**. It forms new indirect connections between $v\dagger$ and v_0 to compensate for their direct connection lost in **Step 1**. In **Step 2**, the degree of $v\dagger$ stays unaffected but the degree of some of its neighbours shall increase which in turn would actually help in hiding the relative importance of $v\dagger$ within the network. The creation of a new link (v_0, v_i) (where v_i is some neighbour of $v\dagger$) cannot increase the closeness centrality of $v\dagger$ beyond its initial state i.e. before executing the ROAM heuristic. This is because the new indirect path $v\dagger \rightarrow v_i \rightarrow v_0$ is certainly longer than the initial direct path $v\dagger \rightarrow v_0$. Similarly, the creation

of this link cannot increase the betweenness centrality of $v\dagger$ beyond its initial state because this is now a longer path. So shortest paths that existed between other nodes via $v\dagger$ has now become longer. Hence, shortest paths might now either avoid $v\dagger$ or remain unchanged but certainly not increase.

On Choosing v_0 and v_is : Based on simulation studies, researchers have chosen v_0 to be the neighbour of $v\dagger$ with the most connections i.e. the second most influential node in the network. v_0 is connected to the $b-1$ neighbours of $v\dagger$ with the least connections(i.e. the $v_i s$). Let us bring in the analogy of connections in Facebook, a popular social media platform. On Facebook like social platform, **Step 1** requires $v\dagger$ to "unfriend" v_0 and **Step 2** requires $v\dagger$ to suggest the friendship of v_0 to the other chosen nodes i.e. $v_i s$. For $v\dagger$ to introduce two individuals to each other, it must first be friends to both. It is to be noted that **Step 2** must be executed before **Step 1** i.e. $v\dagger$ must first introduce v_0 to the other nodes before ending its friendship with v_0 [13].

7.2 Simulation Results in ROAM

We have tried to implement the ROAM heuristic algorithm on the 9/11 terror attack network. The network consists of 36 nodes and 64 edges as shown in the Figure 1. We have used the "networkx"

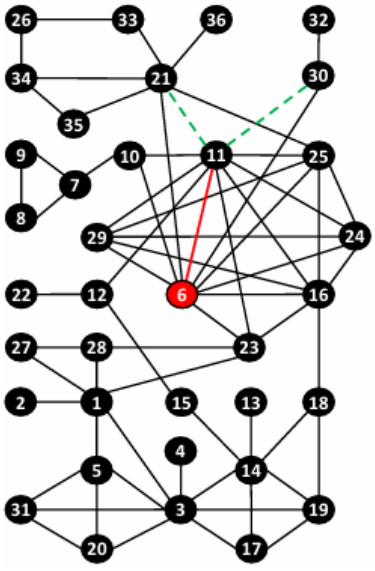


Figure 1: 9/11 terror attack network[13]

package of python for calculating the centrality measures. We used a budget of 3 and executed the ROAM heuristic twice. The mastermind node identified was node 6. The initial and final centrality measures of the mastermind node after two rounds of the heuristic is shown in the following table:

Metric	Initial	Final
Degree	0.257143	0.200000
Closeness	0.426829	0.380435
Betweenness	0.389380	0.109140
Degree Rank	1.000000	4.000000
Closeness Rank	1.000000	4.000000
Betweenness Rank	1.000000	11.000000
Sum of Ranks	3.000000	19.000000

Table 1: Comparison of Initial and Final Centrality Measures of the mastermind node 6

Explanation: The node 6 marked in "red" in Figure 1 is the "mastermind" node. The second most influential node is node 11. In the first round of execution, the edge in red from node 6 to 11 has been removed. Then, two indirect connections as shown by the dotted "green" lines have been formed as edges between nodes $(11, 21)$ and nodes $(11, 30)$. The nodes 21 and 30 were the mastermind's least influential neighbours at the first round of execution. Thus, influence loss due to direct connection removal between nodes $(6, 11)$ has been compensated by the new edges made between nodes 11,21 and nodes 11,30. The plot of the network before and after 2 rounds of ROAM execution has been shown below.

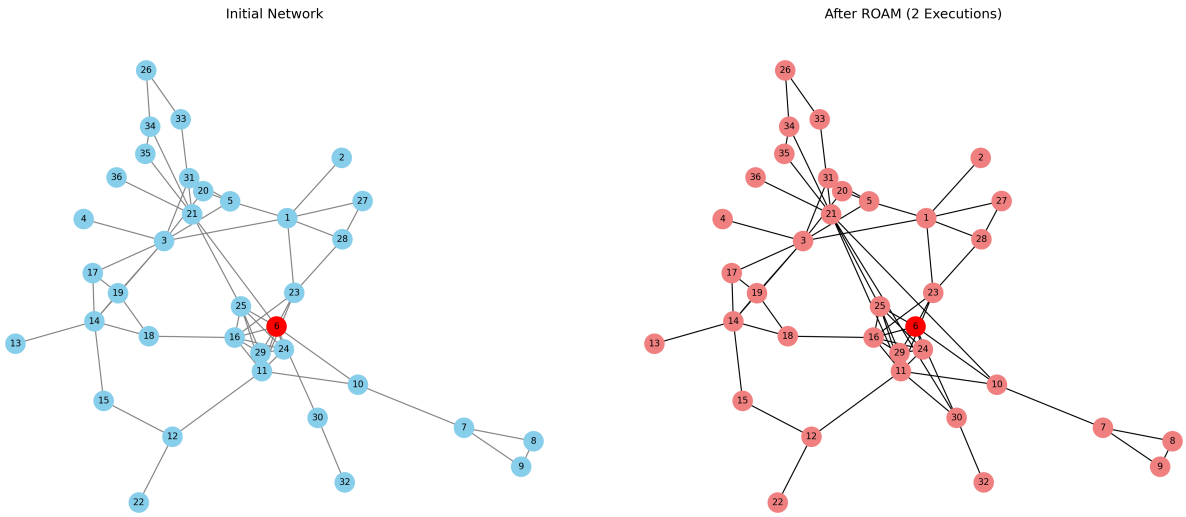


Figure 2: Initial and final network topology

From Table 1 we note that all the centrality measures of the mastermind node have reduced after the heuristic implementation. Naturally, the sum of ranks has also reduced proving the reduction in the node's importance. Hence, the mastermind has been hidden to a considerable extent as compared to its initial state. Now we shall also note the Independent Cascade (IC) and Linear Threshold (LT)

influence values of the mastermind in the table below.

Stage	IC Influence	LT Influence
Initial State	3.596	2.802
After Execution 1	3.252	2.972
After Execution 2	3.118	3.170

Table 2: IC and LT Influence Values of the Mastermind Node

A cascade is the chain reaction of influence spreading through the network. IC(mastermind) is the average number of nodes influenced when the cascade starts from the mastermind. It is the mastermind's direct spreading power. If IC is high, influence spreads widely from the mastermind. Investigators can trace cascade paths backward and easily identify the source (mastermind). If IC is low, spread is limited or less. Cascade is weak and fragmented. Influence does not travel far. Hence the mastermind is harder to trace. In our results, the IC values reduce after each iteration proving that the mastermind is being concealed effectively after each execution of ROAM.

The LT(mastermind) measures how much the network prevents influence to spread from the mastermind. High LT influence indicates high resistance from the nodes i.e. thresholds of each node are high. Hence, influence from a few neighbors is not enough to activate a node. If LT influence is high, nodes would require more collective influence from its peers to activate themselves. Hence, cascades die early. Influence does not spread much which makes it difficult for investigators to trace the mastermind. From our results, we see that LT Influence is increasing after each execution of the heuristic. Hence, we can conclude that ROAM is being able to effectively hide the most influential node in the network.

7.3 DICE - Disconnect Internally, Connect Externally

The next heuristic named DICE is designed to achieve the second objective that we had stated earlier. DICE aims to conceal the identity of a community, $C^{\dagger}$ which exists within a community network structure. A community has been previously defined in our context as a subset of nodes. DICE tries to prevent $C^{\dagger}$ from being identified by community detection algorithms. To hide $C^{\dagger}$, certain links of the network are rewired according to some budget.

Given a budget b, the **DICE heuristic** has the following steps:

Step 1: Disconnect $d \leq b$ links from within $C^{\dagger}$.

Step 2: Connect $b - d$ nodes from within $C^{\dagger}$ to $b - d$ nodes from outside of $C^{\dagger}$.

DICE is inspired by a metric named "modularity". It is used to measure the quality of any given community structure. Modularity promotes structures that have dense connections within communities and sparse connections between them. Community detection algorithms are generally designed to search for a structure that maximizes modularity. **Step 1** of this heuristic decreases the density of the connections within $C^{\dagger}$ and **Step 2** increases the connections between $C^{\dagger}$ and other communities. The heuristic basically tries to decrease the modularity so that community detection algorithms overlook $C^{\dagger}$ i.e. they fail to detect $C^{\dagger}$ and $C^{\dagger}$ stays hidden. This heuristic can be easily applied in platforms like Facebook where **Step 1** would require users to unfriend some people and **Step 2** would require the same users to send friendship requests to some other users who are not already friends with them[13].

7.4 Simulation Results in DICE

We implement the DICE heuristic on the same 9/11 terror network as shown in Figure 1. We take a small dense target community from inside the 9/11 network which comprises these nodes - 6, 11, 16, 23, 24, 25. The heuristic rewires this target community by removing 2 edges from within the community and connecting two edges from within to outside of the community. The removed edges are: [(6, 16), (23, 16)]. The added edges are : [(24, 19), (24, 13)]. Shown below is the plot of the graph before and after execution of DICE. In Table 3, we have reported the modularity value before

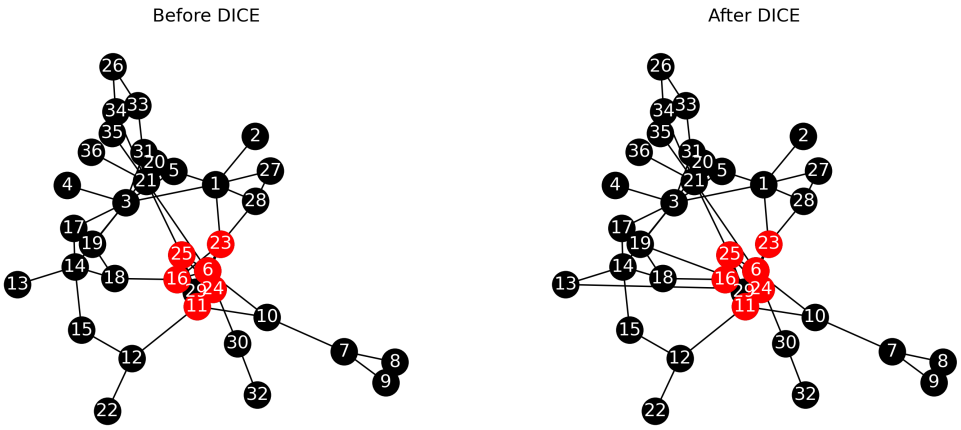


Figure 3: 9/11 terror network before and after DICE

and after execution of the heuristic.

Stage	Modularity
Before	0.2109375
After	0.16748046875

Table 3: Modularity before and after execution of DICE

Interpretation: We can see the drop in modularity value from 0.21 to 0.16. Hence, community detection algorithms which typically detect communities with larger modularity will now have more difficulty in detecting the chosen target community. The target community has now become less structurally distinguishable from the rest of the network. Community boundaries are blurred. The target group is less isolated. Its members look more like the other less important nodes in the network. Hence detection becomes harder.

7.5 HERMES - Hide Exposures by Removing Mastermind's External Sources

We shall first discuss the setup of this heuristic. Previously we have seen state of the art algorithms like ROAM but HERMES takes a slightly different approach to solving the hiding problem. Let $G = (V, E)$ denote a social network, where V is a set of nodes and E being the edges between nodes. A core community $C_i \subseteq V$ is a densely connected subgraph where each node in C_i has a certain minimum degree of connections to other nodes within the same community and they are internally strongly connected. A mastermind $M \in V$ is the most influential node within C_i. In platforms like Facebook, we often see fake news spreading rapidly. In such scenarios, the mastermind spreads misinformations with the help of a core group of influential users who help in spreading and amplifying the fake news. If the mastermind interacts directly with users outside the core group, investigators can easily trace these activities and detect the mastermind. If the direct connection paths are avoided and the mastermind solely relies on the core group to spread the misinformation, then it can stay hidden and detecting it would be difficult. ROAM focuses on the most influential connections of the mastermind across the entire network while HERMES targets a core community that exists within the network[11].

Workflow of the Algorithm : HERMES first identifies a set of communities S_c in G using the Louvain community detection algorithm i.e. the Louvain algorithm splits the community structure into various communities. Next, HERMES ranks each node within each community $C_i \in S_c$ based on

degree centrality(DC), betweenness centrality(BC) and closeness centrality(CC). It is to be noted that these centrality ranks are global ranks considering all the nodes in the network. The average rank of a community is calculated by summing the rank of its nodes and dividing by the total number of nodes in the respective community. The community with the lowest average rank across all three centrality measures is the core community. Next, HERMES finds the most influential node in the core community based on the ranks of DC, CC and BC which have been already calculated. The node with the highest rank (i.e. smallest magnitude) is identified as the mastermind M. If no node ranks best across all the centrality measures, the node with the best rank in the most measures is chosen as the mastermind. After the mastermind is identified, HERMES determines the set of external nodes $NC^{\dagger}$, which are the mastermind's neighbours outside of the core community. From $NC^{\dagger}$, the top d most influential external nodes NC_{max} are selected based on DC, CC and BC values. HERMES operates in iterations selecting the top d external nodes in each iteration. HERMES disconnects these nodes from the mastermind and establishes new edges between these nodes and the less influential nodes of the core community[11].

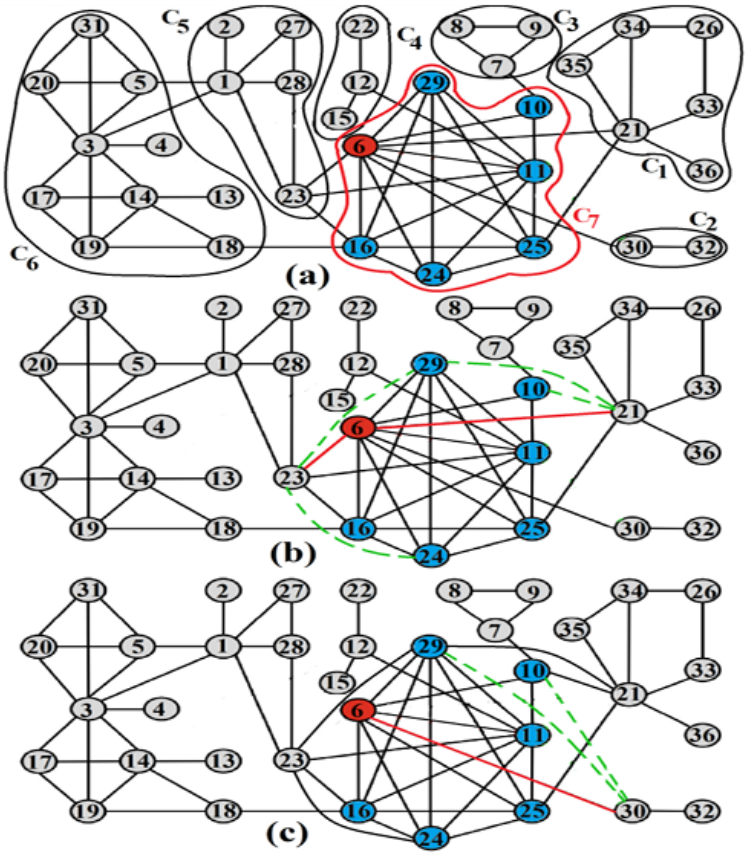


Figure 4: HERMES on 9/11 terror network[11]

Through the above Figure 4, we try to explain the working of the heuristic on the same 9/11 network. Figure 4(a) shows that the network has been divided into 7 communities starting from C_1 to C_7. Community $C_7 = (10, 29, 11, 25, 24, 16, 6)$ has been identified as the core community. Then the heuristic finds node 6 as the mastermind M. Now, a set of external nodes $NC^{\dagger} = (21, 23, 30)$ is chosen which are the mastermind's neighbours outside its own community. From $NC^{\dagger}$, the top d most influential nodes are selected based on centrality. In this example, $d = 2$. The total number of external nodes involved in both connection and disconnection processes is denoted as b, resulting in $b - 1$ new edges being formed(i.e. to keep b nodes connected, we need atleast $b - 1$ edges to form a minimally connected subgraph). In $NC^{\dagger}$, number of nodes is 3 i.e. b = 3 here. In Figure 4(b), in the first iteration, the heuristic identifies 21 and 23 as the top 2 external nodes. It removes the direct connection of the mastermind node 6 with these 2 nodes 21 and 23. The edges in red i.e. $(6, 21)$ and $(6, 23)$ are removed and new links $(21, 29)$, $(21, 10)$ for 21, $(23, 24)$, $(23, 10)$ for 23 are formed. These are shown by green dotted lines. In the next iteration, the heuristic finds node 30 as the top external node remaining inside $NC^{\dagger}$ and removes its connection with the mastermind and the process carries on in the same manner[11].

Algorithm for HERMES heuristic

Considering a graph $G = (V, E)$, the number of executions or iterations denoted as $iter$, the total number of external nodes involved in both connection and disconnection denoted as b and d denoting the number of top external nodes involved in each iteration, the HERMES heuristic has the following steps [11]:

Step 1: Identify the core community C_i in the given graph G.

Step 2: Identify $M \in C_i$ based on the three centrality measures.

Step 3: Determine the set of external nodes $NC^{\dagger}$ connected to M but not in C_i.

Step 4: for k in range(1, iter + 1):

- Find the top d most influential external nodes $NC_{max} \in NC^{\dagger}$ and remove the links between M and the nodes in NC_{max}. Reconnect these nodes with less influential nodes in C_i.
- Remove these d external nodes from $NC^{\dagger}$.

Step 5: end for

7.6 Simulation Results in HERMES

It is to be noted that HERMES cannot perform well in a small network comprising just a few nodes and a few edges. The 9/11 terror network is a small network. So we do not implement HERMES on it. Rather we implement HERMES on relatively larger networks like the 'ego-Facebook' dataset available from SNAP[2][8].The facebook dataset available at SNAP is computationally difficult to execute for an ordinary user since the dataset is huge. It is an undirected network containing 4039 nodes and 88234 edges. The dataset consists of friend lists from Facebook. The computational resource needed to run it is generally not available to a common user. So we fragment the facebook dataset into a smaller version consisting of 333 nodes and 2978 edges. The table below reports the initial and final ranks of the mastermind that we computed in our simulation.

Metric	Initial	Final
Degree Rank	95	101
Closeness Rank	81	203
Betweenness Rank	11	10
Sum of Ranks	187	314

Table 4: Initial vs Final Ranks of Mastermind (Node 2724)

Interpretations : We can see that the sum of ranks has increased after the implementation of the heuristic(i.e. bad rank), indicating that the mastermind node is now less important than before and shall be harder to detect.

Stage	IC	LT
Initial	0.56	0.07
Iteration 1	0.54	0.08
Iteration 2	0.53	0.09

Table 5: IC and LT values across HERMES iterations

From Table 5, we note the trends for the IC and LT values after each iteration of the heuristic. We find that the IC value drops after each iteration and the LT value increases after each iteration. The simulated results are in sync with the theoretical expectations. The mastermind's direct influence over the network has diminished as indicated by the decreasing IC values. The structural resistance of the network towards influence emanating from the mastermind has also strengthened after each iteration as indicated by the rising LT values. Hence influence does not travel far and mastermind becomes more untraceable.

7.7 C3R - Community Constrained Cascade Re-routing

So far, we have studied, analysed and implemented the heuristics that have been researched by researchers across the globe. Now we shall propose our novel heuristic named C3R i.e. Community Constrained Cascade Re-routing. This particular heuristic walks majorly along the line of HERMES in the sense that we approach the problem of hiding from a community perspective. The way we rewire the connections in C3R is what differentiates it from HERMES and makes it an original thought.

Workflow of the Algorithm for C3R

The heuristic involves the following steps :

Step 1: Break down the network into smaller communities using some community detection algorithm (for eg Louvain).

Step 2: Next, each node within each community is ranked based on the degree centrality (DC), closeness centrality (CC), and betweenness centrality (BC).

Step 3: The average rank of a community is determined by summing the ranks of its nodes and dividing by the total number of nodes in that particular community. The community with the lowest average rank across all three centrality measures is the core community.

Step 4: Next, the heuristic finds the most influential node in the core community based on the ranks of DC, CC, and BC. The node with the best rank(i.e. the smallest rank in magnitude across all three centrality measures) is identified as the Mastermind. If no node ranks the best across all three measures, the node with the best rank in most measures is chosen.

Step 5: Once the mastermind is identified, the heuristic determines the set of external nodes, which are the mastermind's neighbours outside the core community.

Step 6: From this set of external nodes, the top d most influential external nodes are selected based on DC ,CC, and BC values. The heuristic operates iteratively, selecting the top d external nodes in each iteration. It then disconnects these nodes from the mastermind and connects these nodes to the weakest available node (Say W_i) in their own respective community (if not connected). Then that node (W_i) is connected to the weakest available node in the core community comprising the mastermind which in turn has to be connected to the mastermind. To say simply, the connection from the mastermind to an outside influential node(say O_i) is being replaced with this new rewiring: Outside influential node(O_i) $\rightarrow$ Weakest node in O_i's community (W_i) $\rightarrow$ Weakest node in mastermind's community

(say MW_i) $\rightarrow$ Mastermind.

The sixth step in C3R is the unique multi-hop rewiring strategy. It re-routes threatful information to flow into the mastermind via the weak nodes of both the outside and the core communities. Naturally, the threat detection capability of the detection mechanisms on information paths passing through multiple weak nodes would be challenged significantly.

7.8 Simulation Results in C3R

We have applied the C3R heuristic on the same fragmented facebook network having 333 nodes and 2978 edges.

Metric	Initial	Final
Degree Rank	91	93
Closeness Rank	89	333
Betweenness Rank	18	14
Sum of Ranks	198	440

Table 6: Initial vs Final Centrality Ranks of Mastermind (Node 2724) for C3R

Stage	IC	LT
Initial	0.6304	0.007
Iteration 1	0.6264	0.010
Iteration 2	0.6089	0.011

Table 7: IC and LT values across C3R iterations

Interpretations: The mastermind node has been identified as the node 2724. The increase in degree rank shows that the mastermind is now slightly less connected locally. The increase in closeness rank shows that the mastermind is now less connected globally. The small improvement in betweenness indicates that the mastermind is now slightly more structurally important. But the overall increase in the sum of ranks shows that the mastermind is now less important in the network and shall be harder to detect via detection techniques. The multi-hop rewiring might have created new bridge paths through the mastermind's community which could have caused the mastermind to lie on more shortest paths, hence improving the betweenness centrality to a small extent.

We also see that the IC value drops after each iteration and the LT value increases after each iteration of the heuristic. These trends are in sync with the theoretical expectations. The direct influence of the mastermind over the network is reducing after each execution of the heuristic. The resistance of the

network towards influence propagating from the mastermind is also increasing after each iteration. Hence influence does not travel far and it becomes difficult for investigators to trace the mastermind.

Stage	HERMES IC	HERMES LT	C3R IC	C3R LT
Initial	0.56	0.07	0.6304	0.007
Iteration 1	0.54	0.08	0.6264	0.010
Iteration 2	0.53	0.09	0.6089	0.011

Table 8: Comparison of IC and LT Values for HERMES and C3R

From Table 8, we can see that the value of IC at each iteration for HERMES is lower than that of C3R. Also, the value of LT at each iteration for HERMES is higher than that of C3R. So we can say that our novel heuristic is slightly weaker than HERMES on the facebook dataset, in terms of reducing the influence of the mastermind in the form of cascades. But it very well matches the theoretical expectations for the facebook dataset. Also we can evaluate the actual performance of C3R when we test this heuristic over multiple networks. We can then have an averaged assessment of its performance.

8 Conclusion

Throughout the course of this work, we have tried our best to simulate the various heuristics and to see their results on some available datasets. We have used the 9/11 dataset and the facebook fragmented dataset. In order to evaluate the heuristics more precisely and to validate their results, we must simulate them on a wide range of data. Aside from the entire facebook dataset comprising 4039 nodes and 88234 edges, we also have social circle data from Google+ available at SNAP. But that is a directed network. Accordingly, changes have to be made in the code. We must also recognize that sophisticated computational resource is needed to run these algorithms on such huge datasets. In our IC model, each edge has a fixed probability $p = 0.15$. It means that each active node gets one chance to activate each of its neighbours with probability 0.15. As a research investigation, we can take a grid value of this probability and note the variation in the trend of the IC values. In the linear threshold function, we use $delta = 0.05$. It is a resistance increment parameter which implies that every time a node fails to get activated, its threshold is increased by 0.05. In the standard LT model, threshold is fixed. In our version, we have tried to make the model more realistic by implementing this parameter. This is in sync with the nature of adversarial networks where nodes do not easily change behaviour and repeated failed attempts in activating a node ends up in making it more resilient. The community detection algorithm that we used in our work is louvain algorithm. We can experiment with different choices of community detection algorithms to see their impact on the final results. Community detection algorithms like Girvan Newman and Label Propagation algorithms can be used.

The hiding problem that we addressed in this report has immense future scopes. Researchers can try to develop better heuristics. One can also venture into the problem of hiding where an individual tries to hide in a multilayer connection of networks. In such a case, the individual is present in multiple social networks simultaneously. One can also try to solve the link prediction problem. Link prediction problem deals with predicting future possible links in the network based on the current state of the network.

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